

YIWEI WANG

☎ 626-492-6763

✉ yiweiwang2007@gmail.com

🔗 yiweipwang.github.io

Education

Brown University

Expected May 2029

B.S. in Applied Mathematics-Computer Science, GPA: 4.00/4.00

Providence, RI

Relevant Coursework: Linear Algebra, Partial Differential Equations, Statistics and Probability, Data Structures and Algorithms, Operations Research: Probabilistic Models

Activities: Brown University Math Olympiad Problem Writer, Quant Trading at Brown Member

Honors and Awards

IMC Prosperity 4 Finalist

3× AIME Qualifier (Score: 12/15)

4× Los Angeles Math Field Day Gold Medalist

Coca-Cola Scholar Semifinalist (Top 1.27% of 105,000)

Skills

Languages: Python, R, C++, JavaScript, MATLAB, CSS

Libraries/Tools: JavaFX, Pandas, NumPy, Matplotlib

Technical: Data Analysis, Statistical Modeling, Data Visualization, Algorithm Design

Work Experience

Icarus Fund

June 2026 – Present

Quantitative Research Intern

New York City, NY

- Engineered a **cross-sectional alpha factor framework** across **50+ signals** on S&P 500, performing **IC/ICIR** analysis and **t-stat** screening to identify high-conviction factors for model inclusion.
- Built a **multi-stage feature selection pipeline** using rank correlation matrices and VIF analysis to eliminate redundant factors, reducing the signal universe while maximizing predictive power for systematic equity ranking.
- Developing a **regime-aware portfolio construction system** integrating **XGBoost** regression/ranking models with a **Hidden Markov Model** for market classification to dynamically adjust exposure and position sizing.

Polar Education

June 2023 – Present

Chief Operating Officer

La Canada Flintridge, CA

- Manage **\$100K+ in operational budgets**, overseeing payroll, financial allocation, and staff coordination for **tutoring organization of 20+ members**.
- Deliver **400+ hours** of one-on-one tutoring in math, SAT preparation, and computer science, achieving **95%+ client retention rate**.
- Communicate with schools nationwide to design **tailored lesson plans**, aligning pedagogy and assessment strategies to maximize student outcomes.

Research Experience

Smartphone Use on Chronic Health

June 2024 – August 2024

Research Assistant | Advisor: Prof. David Gatchell

Northwestern University

- Engineered data pipelines** to clean health datasets from **15+ peer-reviewed studies**, enabling systematic analysis of screen exposure metrics.
- Applied **statistical modeling** techniques across cardiometabolic and biometric datasets to isolate key **predictors** of sleep disruption.
- Collaborated** with a **4-person research team** to validate **data driven intervention models**, iterating on findings to improve predictive accuracy of behavioral outcomes.

Computational Combinatorics and Algorithms

June 2023 – October 2023

Research Assistant | Advisor: Prof. Greg Dresden

Washington and Lee University

- Developed **recursive algorithms** to **enumerate tiling configurations** of hexagonal boards, extending **Fibonacci sequence theory** through mathematical modeling of trapezoidal tile arrangements.
- Derived novel relations integrating **Narayana's Cow sequence** to reduce computational complexity from **exponential to polynomial time**.
- Published integer sequence **A365352** in the **OEIS** and achieved **top 8.9%** ranking in *Pioneer Research Journal* submissions.

Project Experience

Minesweeper with AI System | Java, JavaFX

December 2025

- Implemented a Minesweeper application in **JavaFX**, including event driven UI, and scalable board configuration.
- Implemented **BFS flood-fill** reveal mechanics to automatically expand safe regions and improve playability.
- Designed a **probability-based AI** using a **binary tree risk model**, achieving **96% win rate** on **500 test boards**.